# Supporting Information for: "A Universal Quantum Descriptor for Electron Correlation in Molecular Systems: The $F_{\text{bond}}$ Framework"

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# 1 Complete Computational Details

## 1.1 Software and Hardware

All calculations were performed using:

- **Software:** PySCF v2.X [1, 2]
- **Hardware:** AMD Ryzen 9 9950X (16 cores @ 3.5 GHz), 64 GB RAM
- **Operating System:** Linux (Ubuntu 24.04)

## 1.2 Basis Set Details

**STO-3G:** Minimal basis set

- H: 1 basis function (1s)
- C: 5 basis functions (1s, 2s, $2p_x$, $2p_y$, $2p_z$)
- N: 5 basis functions (1s, 2s, $2p_x$, $2p_y$, $2p_z$)
- O: 5 basis functions (1s, 2s, $2p_x$, $2p_y$, $2p_z$)

**6-31G:** Split-valence basis set (used for $H_2O$ validation)

- H: 2 basis functions
- O: 13 basis functions

## 1.3 Frozen-Core Treatment

For molecules containing carbon, nitrogen, or oxygen, core orbitals were frozen at the Hartree-Fock level:

Table 1: Frozen-core configuration for each molecule.

| Molecule | Core Orbitals | Core Electrons | Active Space |
|---|---|---|---|
| $H_2$ | None | 0 | (2e, 2o) |
| $NH_3$ | N 1s | 2 | (8e, 8o) |
| $H_2O$ | O 1s | 2 | (8e, 6o) |
| $CH_4$ | C 1s | 2 | (8e, 8o) |
| $C_2H_4$ | 2× C 1s | 4 | (12e, 13o) |
| $C_2H_2$ | 2× C 1s | 4 | (10e, 11o) |
| $N_2$ | 2× N 1s | 4 | (10e, 10o) |

**Rationale:** Core electrons (1s orbitals) do not participate in chemical bonding and their correlation with valence electrons can create artificial fractional occupations in minimal basis sets. Frozen-core treatment ensures $F_{\text{bond}}$ $measuresonlychemicalcorrelationrelevanttobonding.$

## 1.4 Convergence Criteria

- **Hartree-Fock:** SCF convergence threshold $10^{-10}$ Ha
- **CCSD:** Energy convergence $10^{-8}$ Ha, amplitude convergence $10^{-7}$
- **FCI:** CI vector convergence $10^{-10}$, implemented via CASCI with complete active space

## 2 Molecular Geometries

All geometries are provided in Ångstroms. Coordinates for experimentally known molecules were taken from NIST or optimized at B3LYP/6-311G** level.

### 2.1 Hydrogen ($H_2$)

```
H  0.0000  0.0000  0.0000
H  0.0000  0.0000  0.7414
```

Bond length: 0.7414 Å (equilibrium)

### 2.2 Nitrogen ($N_2$)

```
N  0.0000  0.0000  0.0000
N  0.0000  0.0000  1.0980
```

Bond length: 1.098 Å (equilibrium)

### 2.3 Ammonia ($NH_3$)

```
N  0.0000  0.0000  0.1160
H  0.0000  0.9377 -0.2715
H  0.8117 -0.4689 -0.2715
H -0.8117 -0.4689 -0.2715
```

N-H bond length: 1.012 Å, H-N-H angle: 106.7°

### 2.4 Water ($H_2O$)

```
O  0.0000  0.0000  0.1173
H  0.0000  0.7572 -0.4692
H  0.0000 -0.7572 -0.4692
```

O-H bond length: 0.958 Å, H-O-H angle: 104.5°

### 2.5 Methane ($CH_4$)

```
C  0.0000  0.0000  0.0000
H  0.6276  0.6276  0.6276
H -0.6276 -0.6276  0.6276
H -0.6276  0.6276 -0.6276
H  0.6276 -0.6276 -0.6276
```

C-H bond length: 1.087 Å (tetrahedral, $T_d$ symmetry)

### 2.6 Ethylene ($C_2H_4$)

```
C  0.0000  0.0000  0.6695
C  0.0000  0.0000 -0.6695
H  0.0000  0.9289  1.2321
H  0.0000 -0.9289  1.2321
H  0.0000  0.9289 -1.2321
H  0.0000 -0.9289 -1.2321
```

C=C bond length: 1.339 Å, C-H bond length: 1.086 Å (planar, $D_{2h}$ symmetry)

### 2.7 Acetylene ($C_2H_2$)

```
C  0.0000  0.0000  0.6010
C  0.0000  0.0000 -0.6010
H  0.0000  0.0000  1.6630
H  0.0000  0.0000 -1.6630
```

CC bond length: 1.202 Å, C-H bond length: 1.062 Å (linear, $D_{\infty h}$ symmetry)

## 3 Complete Natural Orbital Analysis

### 3.1 Hydrogen ($H_2$)

Table 2: Natural orbital occupations for $H_2$ (STO-3G, full FCI).

| Orbital | Occupation | Deviation | Entropy (nats) |
|---|---|---|---|
| NO 1 () | 1.9760 | 0.0240 | 0.0681 |
| NO 2 (*) | 0.0240 | 0.0240 | 0.0667 |

### 3.2 Methane ($CH_4$)

Table 3: Natural orbital occupations for $CH_4$ (STO-3G, frozen-core FCI).

| Orbital | Occupation | Deviation | Entropy (nats) |
|---|---|---|---|
| NO 1 | 1.9852 | 0.0148 | 0.0436 |
| NO 2 (C-H) | 1.9761 | 0.0239 | 0.0647 |
| NO 3 (C-H) | 1.9761 | 0.0239 | 0.0647 |
| NO 4 (C-H) | 1.9761 | 0.0239 | 0.0647 |
| NO 5 (*) | 0.0222 | 0.0222 | 0.0611 |
| NO 6 (*) | 0.0222 | 0.0222 | 0.0611 |
| NO 7 (*) | 0.0222 | 0.0222 | 0.0611 |
| NO 8 (*) | 0.0197 | 0.0197 | 0.0554 |

### 3.3 Ethylene ($C_2H_4$)

Table 4: Natural orbital occupations for $C_2H_4$ (STO-3G, frozen-core FCI).

| Orbital | Occupation | Deviation | Entropy (nats) |
|---|---|---|---|
| NO 1-5 () | 1.974-1.984 | 0.016-0.026 | 0.046-0.069 |
| NO 6 () | 1.8823 | 0.1177 | 0.2238 |
| NO 7 (*) | 0.1181 | 0.1181 | 0.2243 |
| NO 8-13 (*) | 0.010-0.026 | 0.010-0.026 | 0.030-0.068 |

**Note:** /* orbitals show strong fractional occupations (11.8% depletion/population), characteristic of -* correlation.

Table 5: Natural orbital occupations for $C_2H_2$ (STO-3G, frozen-core FCI).

| Orbital | Occupation | Deviation | Entropy (nats) |
|---|---|---|---|
| NO 1-3 () | 1.977-1.981 | 0.019-0.023 | 0.053-0.061 |
| NO 4 ($_x$) | 1.9158 | 0.0842 | 0.1745 |
| NO 5 ($_y$) | 1.9158 | 0.0842 | 0.1745 |
| NO 6 ($^*_x$) | 0.0843 | 0.0843 | 0.1747 |
| NO 7 ($^*_y$) | 0.0843 | 0.0843 | 0.1747 |
| NO 8-10 (*) | 0.010-0.031 | 0.010-0.031 | 0.029-0.081 |

### 3.4 Acetylene ($C_2H_2$)

**Note:** Two degenerate  systems show identical fractional occupations (8.4%).

## 4 Computational Timings

Table 6: Typical runtimes on AMD Ryzen 9 9950X (16 cores).

| Molecule | Basis | HF (s) | CCSD (s) | FCI (s) |
|---|---|---|---|---|
| $H_2$ | STO-3G | <1 | 1 | 1 |
| $CH_4$ | STO-3G | <1 | 5 | 10 |
| $N_2$ | STO-3G | 1 | 10 | 15 |
| $NH_3$ | STO-3G | <1 | 8 | 12 |
| $H_2O$ | 6-31G | <1 | 10 | 15 |
| $C_2H_4$ | STO-3G | 1 | 45 | 60 |
| $C_2H_2$ | STO-3G | 1 | 8 | 12 |

**Note:** $C_2H_4$ requires longer time due to larger active space (12e, 13o).

## 5 Validation: CCSD vs. FCI Energy Comparison

Table 7: Energy differences between frozen-core CCSD and FCI.

| Molecule | $\lvert E_{\mathrm{FCI}} - E_{\mathrm{CCSD}} \rvert$ (mHa) | Agreement |
|---|---|---|
| $CH_4$ | 0.23 | Excellent |
| $C_2H_4$ | 1.22 | Excellent |
| $C_2H_2$ | 2.49 | Excellent |
| $N_2$ | 4.02 | Good |

Small differences ($< 5$ mHa) validate that frozen-core treatment is consistent across CCSD and FCI methods.

## 6 Basis Set Sensitivity

For $H_2O$, we tested basis set dependence:

**Key finding:** Classification remains consistent (mean-field) across basis sets, though quantitative values change. STO-3G provides reliable qualitative predictions.

Table 8: Basis set comparison for $H_2O$.

| Basis | $O_{MOS}(Ha)$ | $S_{E,\max}(nats)$ | $F_{bond}$ | Classification |
|---|---|---|---|---|
| STO-3G | 1.120 | 0.0009 | $8.5 \times 10^{-4}$ | Mean-field |
| 6-31G | 0.705 | 0.0027 | $9.6 \times 10^{-4}$ | Mean-field |

# 7 Code Availability

All Jupyter notebooks used for calculations are available as supplementary files:

- `CH4_Frozen_Core_STO3G.ipynb`
- `C2H4_Frozen_Core_STO3G.ipynb`
- `C2H2_Frozen_Core_STO3G.ipynb`

Each notebook contains complete, executable code for reproducing results.

# 8 References

## References


[1] Sun, Q.; et al. PySCF: the Python-based simulations of chemistry framework. *WIREs Comput. Mol. Sci.* **2018**, *8*, e1340.

[2] Sun, Q.; et al. Recent developments in the PySCF program package. *J. Chem. Phys.* **2020**, *153*, 024109.